

Introduction to Microeconomics

Lecture 16—Profit Maximization, Supply and Market Equilibrium in Perfectly Competitive Markets

Prof. Alejandro Melo-Ponce



NAZARBAYEV
UNIVERSITY

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Revenues

Given a market demand curve that a firm faces $Q = D(p)$ with inverse demand curve $p = D^{-1}(Q)$, revenues for the firm are defined as:

$$R(Q) = \text{Price} \times \text{Quantity} = D^{-1}(Q) \times Q.$$

For a linear market demand curve $Q = a - Bp$, the inverse demand is $p = a/b - Q/b$, thus

$$R(Q) = \left(\frac{a}{b} - \frac{Q}{b}\right) Q$$

Profit Maximization

The firm's profit function, for any Q , is

$$\Pi(Q) = R(Q) - C(Q)$$

where $R(Q)$ are revenues and $C(Q)$ are costs as a function of quantity.

The firm's problem is then to **maximize profits** by choosing optimally the quantity offered.

$$\max_Q R(Q) - C(Q)$$

The optimality condition for this problem is:

$$MR(Q^*) = MC(Q^*)$$

Perfectly Competitive Market

A perfectly competitive market is characterized by

- No barriers to entry.
- Homogeneous product.
- Many small firms compared to the total size of the market.
- Firm acts as a price taker.

We will develop the theory of the firm operating in a perfectly competitive market. However, many of the insights that we develop here extends to cases in which some of the characteristics above are not true. Turns out that none of the above are essential for a market to be “competitive”. What is important is that the market has a **contestable structure: low or nonexistent sunk costs, and the possibility to use strategies to undercut rivals, i.e., price adjustments by the firms are felt strongly by consumers.**